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Meng

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(54) **PLUG-IN GROUND LIGHT**

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(51) **Int. Cl.**

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F21V 17/12	(2006.01)
F21V 21/08	(2006.01)
F21V 23/00	(2015.01)
F21V 31/00	(2006.01)
F21Y 115/10	(2016.01)

(52) **U.S. Cl.**
CPC ***F21V 15/01*** (2013.01); ***F21S 6/005*** (2013.01); ***F21V 17/12*** (2013.01); ***F21V 21/0824*** (2013.01); ***F21V 23/002*** (2013.01); ***F21V 31/005*** (2013.01); *F21Y 2115/10* (2016.08)

(58) **Field of Classification Search**
CPC F21S 6/005–008; F21V 15/01–015; F21V 17/12–168; F21V 21/0824; F21V 23/001–002; F21V 31/00–005; F21Y 2115/10
See application file for complete search history.

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315/149

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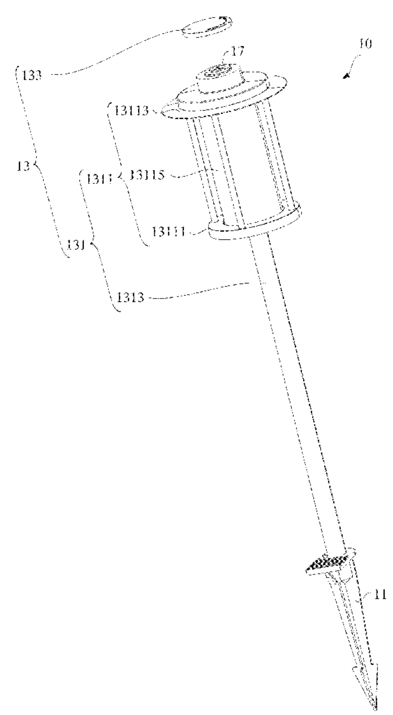
(57) **ABSTRACT**

Provided is a plug-in ground light, including a ground plug pin, a light pole assembly, a light-emitting module, and a functional member. The light pole assembly includes a mounting assembly connected to the ground plug pin and a protective cover connected to the mounting assembly. The protective cover has an open state and a closed state relative to the mounting assembly. The light-emitting module is disposed in the mounting assembly. The functional member is disposed in the mounting assembly. When the protective cover is in the open state, the functional member is exposed, and when the protective cover is in the closed state, the protective cover obscures the functional member. The functional member is configured to control at least one of an operating state, luminous intensity, a luminous color, and an operating mode of the light-emitting module.

15 Claims, 8 Drawing Sheets



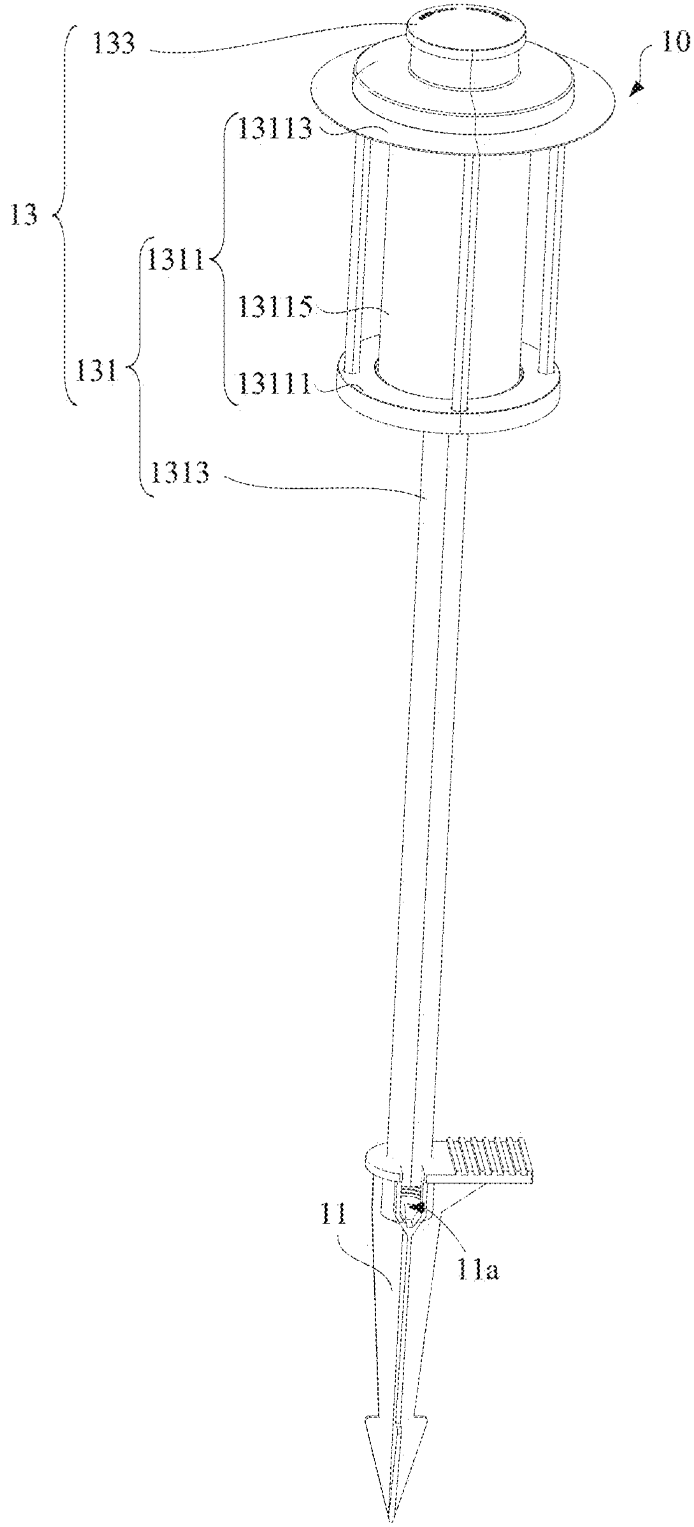


FIG. 1

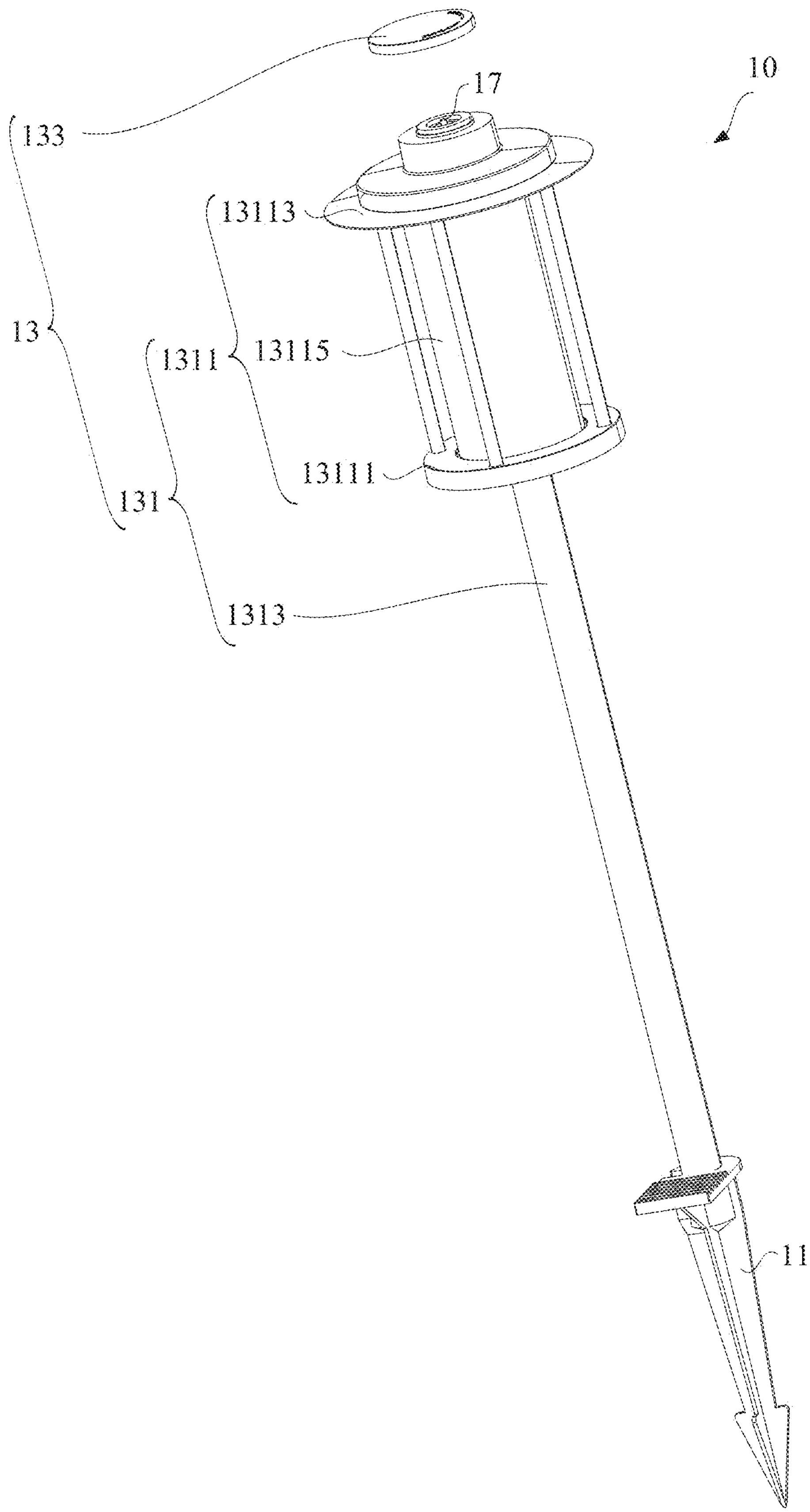


FIG. 2

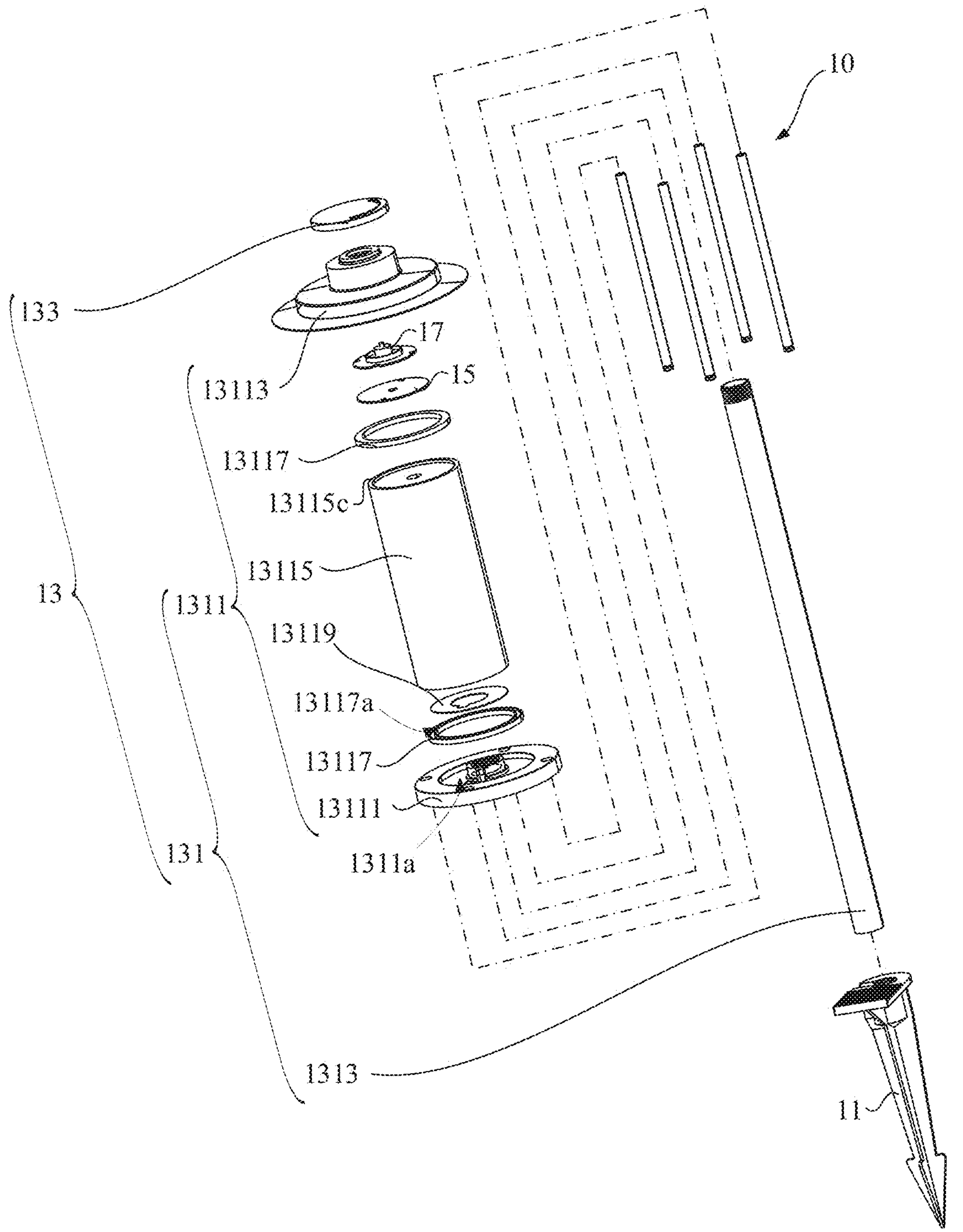
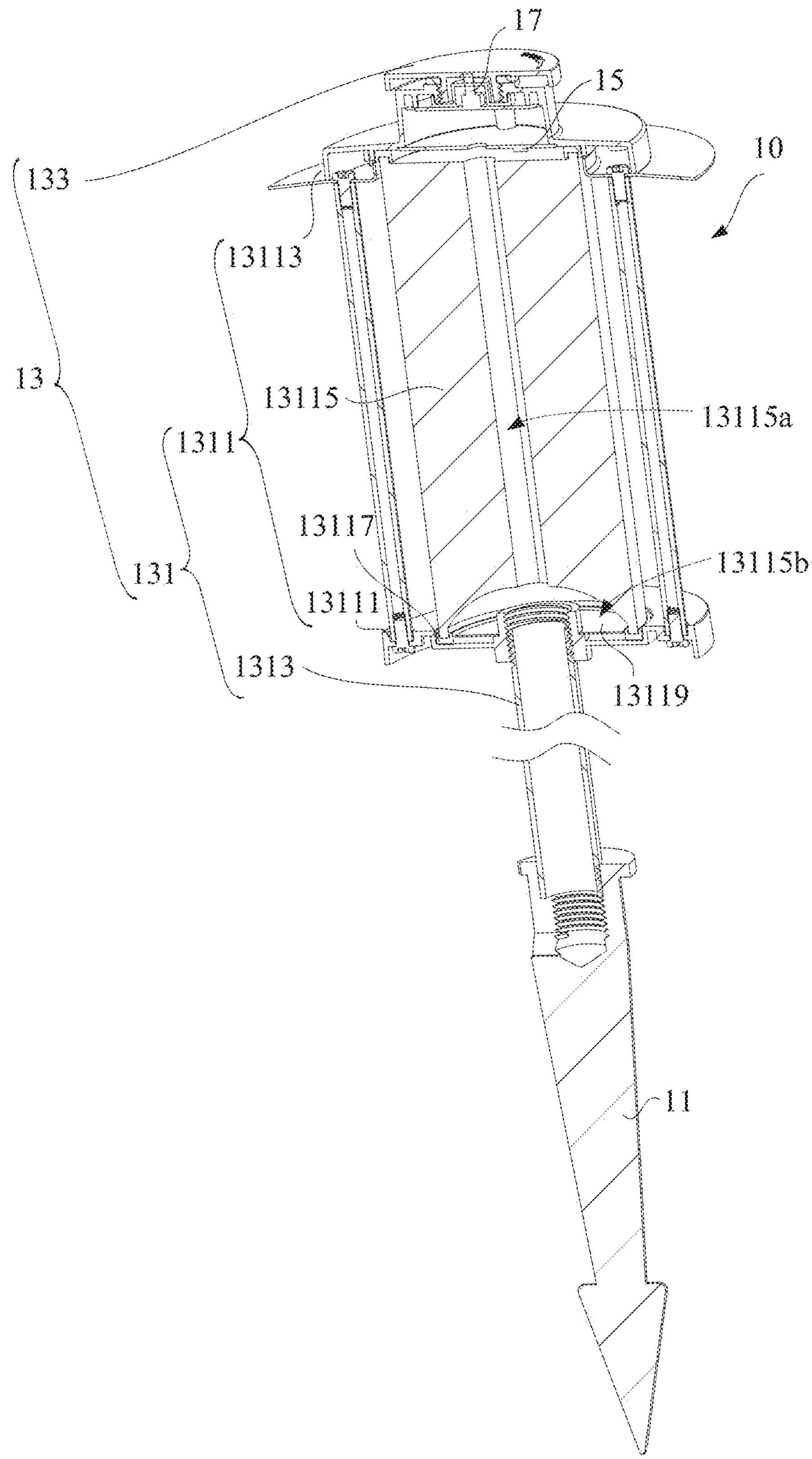

10
133
17
13113
15
13117
13115c
13115
13
1311
13119
13117a
13117
13111
131
1311a
1313
11

FIG. 3


17
15
10
133
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13
13115
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1311
13117
131
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13119
11

FIG. 4

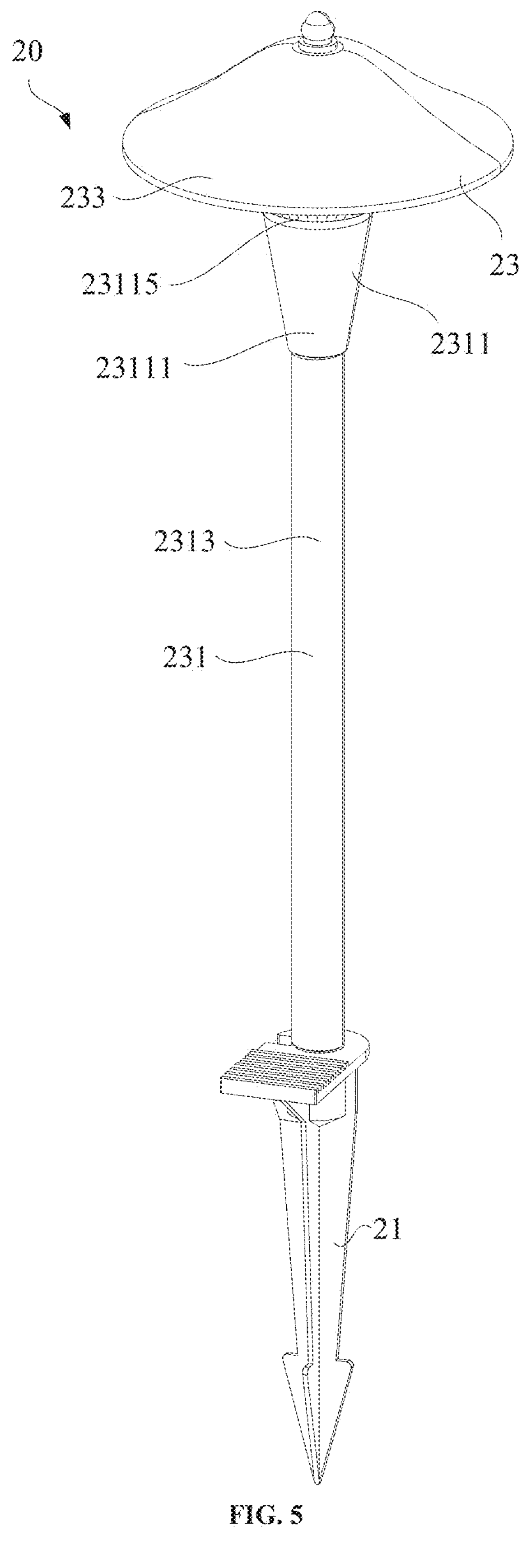

FIG. 5

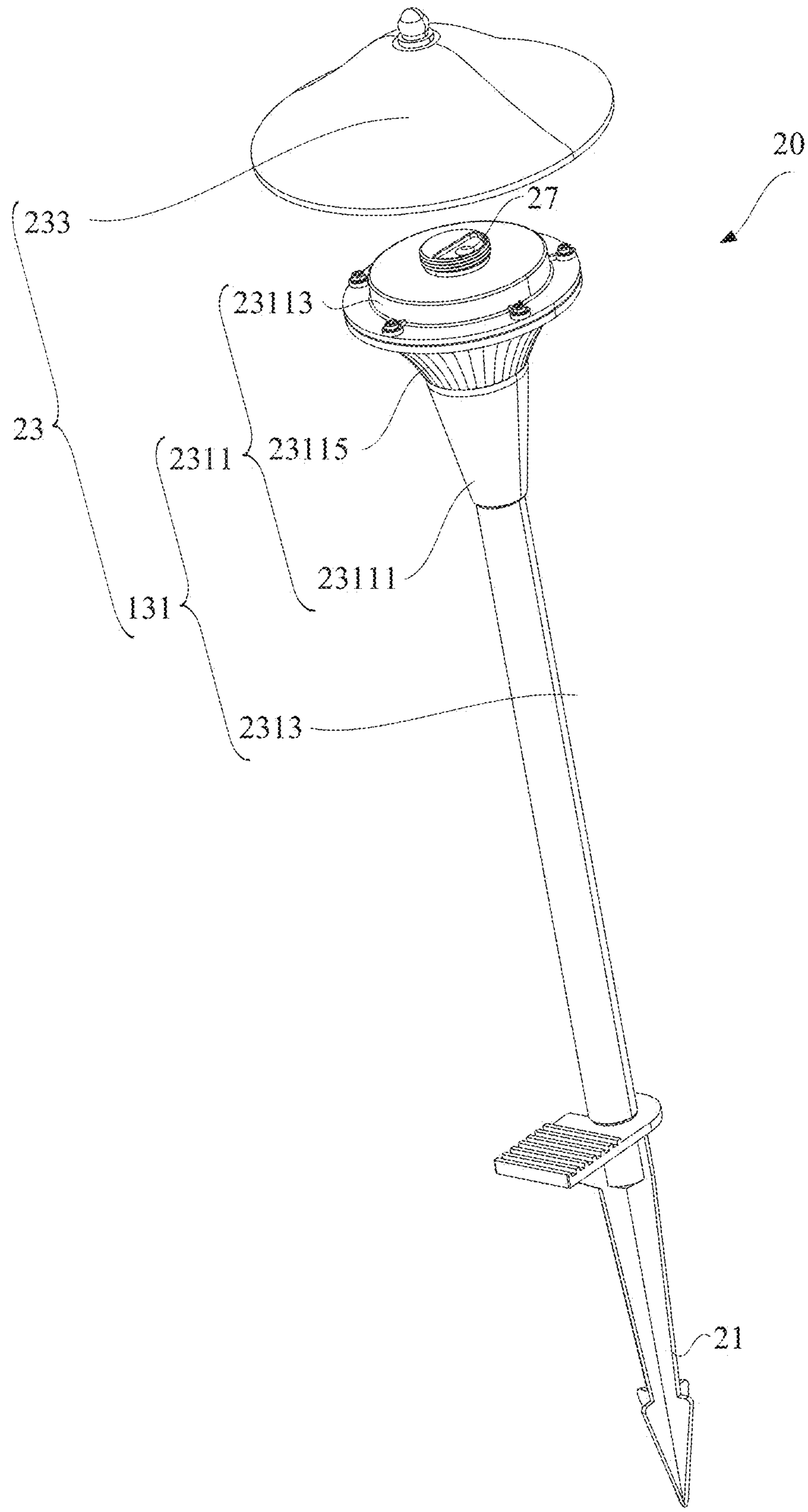

FIG. 6

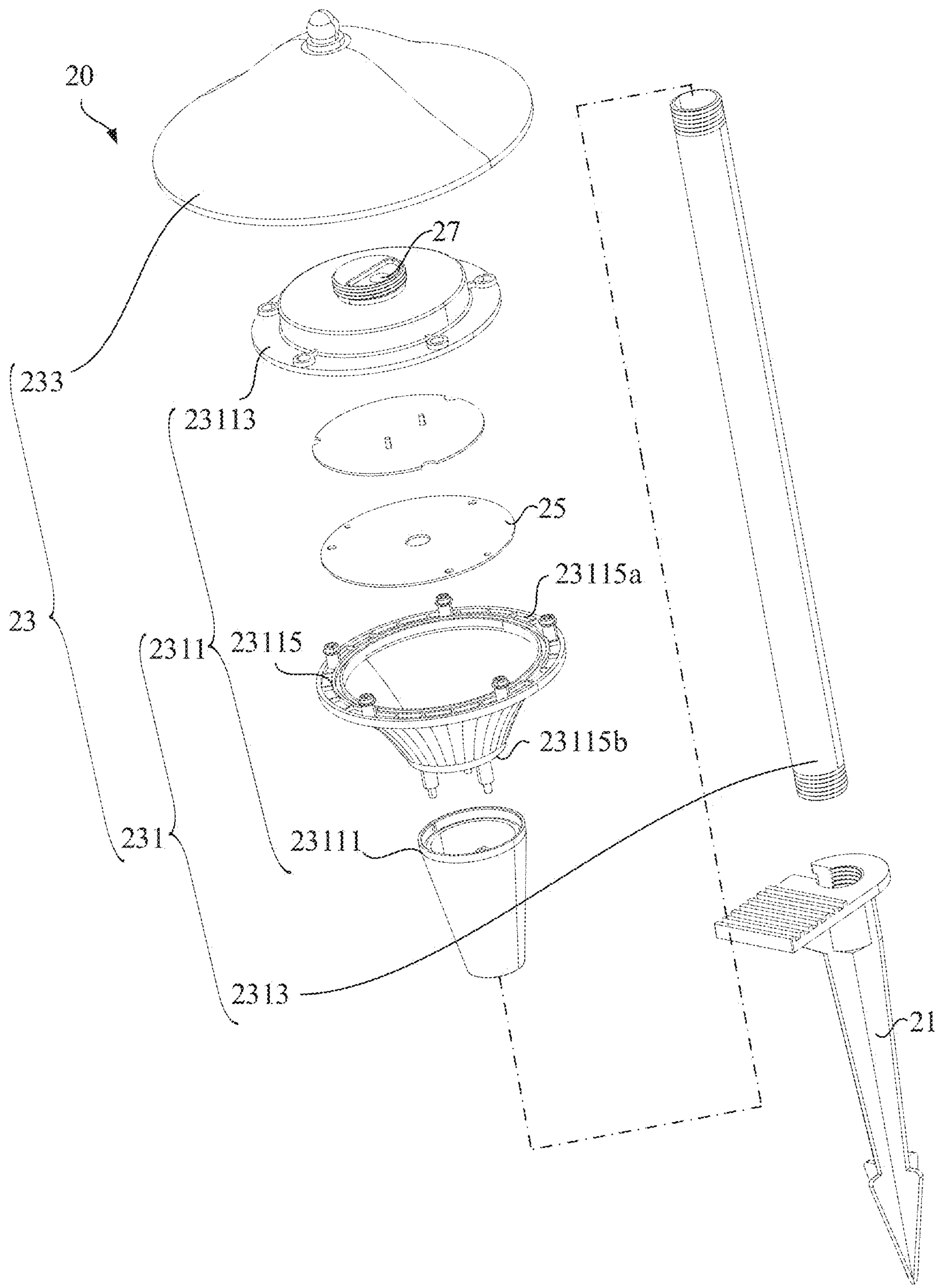
20
27
233
23113
25
23115a
23
2311
23115
23115b
231
23111
2313
21

FIG. 7

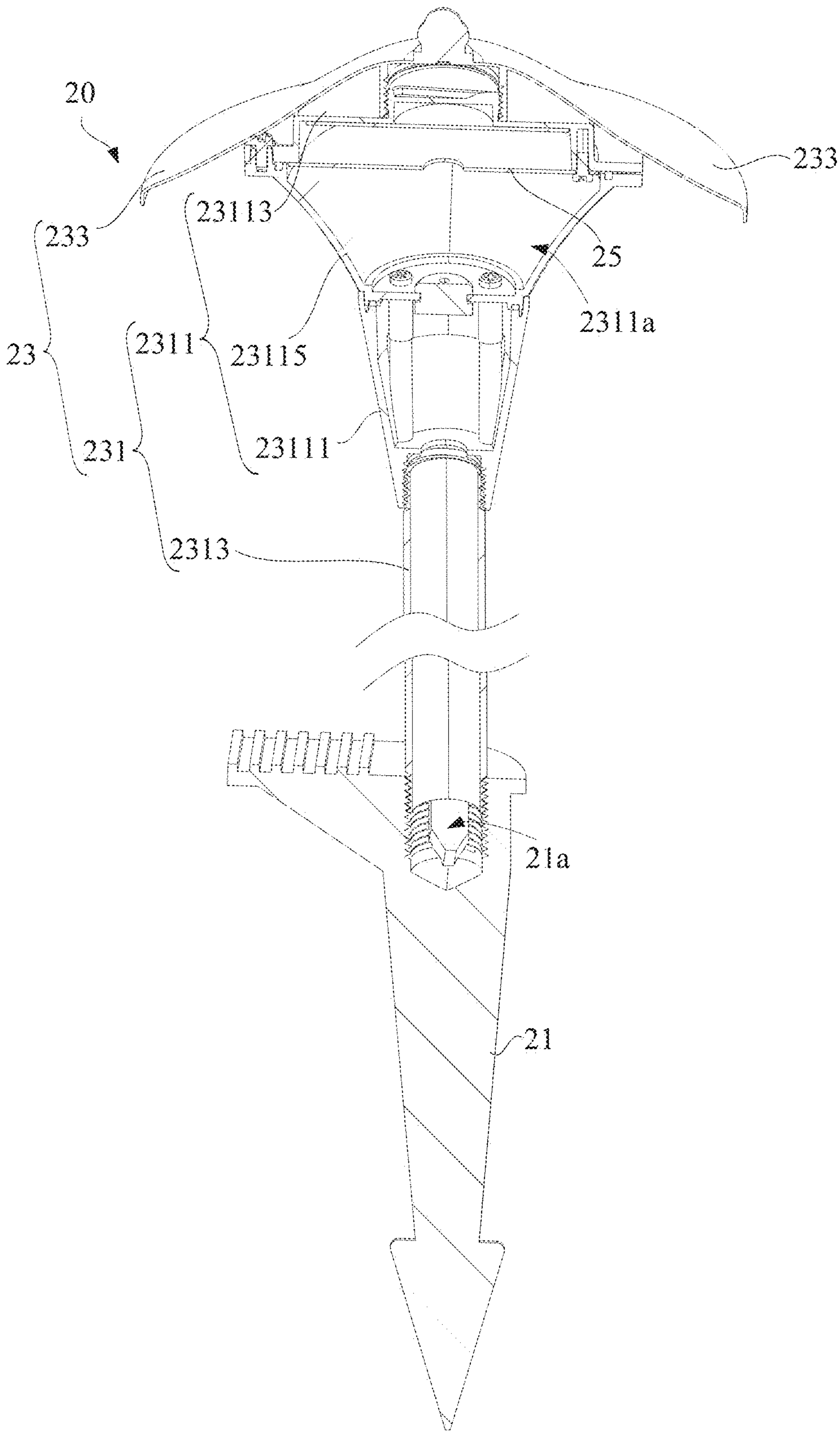

FIG. 8

PLUG-IN GROUND LIGHT

TECHNICAL FIELD

The present disclosure relates to the technical field of lights, and in particular to a plug-in ground light.

BACKGROUND

As an outdoor lighting device, the plug-in ground light is usually mounted on the ground, to provide lighting for places such as roads, parks, and courtyards. In related technologies, the single plug-in ground light usually has fixed lighting patterns such as the fixed illumination brightness, which greatly limits the application scenarios of the plug-in ground light.

SUMMARY

Embodiments of the present disclosure provide a plug-in ground light, to expand a usage scene of the plug-in ground light.

A plug-in ground light includes:

a ground plug pin;

a light pole assembly, including a mounting assembly connected to the ground plug pin and a protective cover connected to the mounting assembly, where the protective cover has an open state and a closed state relative to the mounting assembly;

a light-emitting module, disposed in the mounting assembly; and

a functional member, disposed in the mounting assembly, where when the protective cover is in the open state, the functional member is exposed, when the protective cover is in the closed state, the protective cover obscures the functional member, the functional member is configured to control at least one of an operating state, luminous intensity, a luminous color, and an operating mode of the light-emitting module.

BRIEF DESCRIPTION OF THE DRAWINGS

To describe the technical solutions in the embodiments of the present disclosure or in the prior art more clearly, the following briefly describes the accompanying drawings required for describing the embodiments or the prior art. Apparently, the accompanying drawings in the following description show merely some embodiments of the present disclosure, and a person of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts.

FIG. **1** is a schematic diagram of a protective cover of a plug-in ground light in a closed state according to Embodiment 1 of the present disclosure;

FIG. **2** is a schematic diagram of a protective cover of a plug-in ground light in an open state according to Embodiment 1 of the present disclosure;

FIG. **3** is an exploded view of a plug-in ground light according to Embodiment 1 of the present disclosure;

FIG. **4** is a sectional view of a plug-in ground light according to Embodiment 1 of the present disclosure;

FIG. **5** is a schematic diagram of a protective cover of a plug-in ground light in a closed state according to Embodiment 2 of the present disclosure;

FIG. **6** is a schematic diagram of a protective cover of a plug-in ground light in an open state according to Embodiment 2 of the present disclosure;

FIG. **7** is an exploded view of a plug-in ground light according to Embodiment 2 of the present disclosure; and

FIG. **8** is a sectional view of a plug-in ground light according to Embodiment 2 of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Detailed description of the preferred embodiments is a preferred mode of implementation of the present disclosure. The description has no limitation, and is intended to illustrate the general principles of the present disclosure.

It should be understood that, in order to facilitate the understanding of the present disclosure, the terms "mounting", "connection", "coupling", and "installation" in the following description refer to the connection relationships in the accompanying drawings. For example, "connection" may be understood as a permanent connection or a detachable connection. In addition, "connection" may also be understood as a direct connection, an indirect connection, or a connection implemented via other auxiliary components. Therefore, the above terms should not be a limitation to an actual connection of the elements in the present disclosure.

It should be noted that, the terms such as "length", "width", "top", "bottom", "front", "rear", "left", "vertical", "horizontal", "upper", "lower", "outer", and "inner" are used to indicate the orientation or position relationship in the accompanying drawings. To facilitate the understanding of the present disclosure, an actual position or orientation in the present disclosure is not limited. Therefore, the above terms are not limited to the actual positions of all components in the present disclosure.

It should be understood that the terms such as "first", "second", "a", "an", and "one" in the embodiments refer to "at least one" or "one or more". Specially, the term "a" may refer to "one" in an embodiment and "more than one" in another embodiment. Therefore, the above terms are not limited to actual figures of the components in the present disclosure.

To facilitate the understanding of the present disclosure, the present disclosure is described more completely below with reference to the accompanying drawings. Preferred embodiments of the present disclosure are shown in the drawings. However, the present disclosure is embodied in various forms without being limited to the embodiments set forth herein. On the contrary, these embodiments are provided to make the understanding of the present disclosure more thorough and comprehensive.

Embodiment 1

Referring to FIG. **1**, FIG. **2**, and FIG. **3**, Embodiment 1 of the present disclose provides a plug-in ground light **10**. The plug-in ground light **10** includes a ground plug pin **11**, a light pole assembly **13**, a light-emitting module **15**, and a functional member **17**. The light pole assembly **13** is connected to the ground plug pin **11**, and the light-emitting module **15** and the functional member **17** are disposed in the light pole assembly **13**. One end of the ground plug pin **11** is relatively sharp and may be inserted into the ground, thereby reliably fixing the plug-in ground light **10** on the ground for providing lighting. The light-emitting module **15** may include a light-emitting diode (LED) bead. The LED bead may emit light after being energized. In some implementations, the light-emitting module **15** may include a plurality of LED beads that may emit different colors of light. For example, the light-emitting module **15** includes at least two of a red LED bead, a blue LED bead, a green LED bead, and a white LED bead.

Certainly, the plug-in ground light **10** may include a circuit board. The circuit board is disposed in the light pole assembly **13**, and a control circuit of the plug-in ground light **10** may further be integrated with the circuit board. The control circuit may control a working state of the light-emitting module **15**, for example, turning on or off, controlling luminous intensity of the light-emitting module **15**, controlling a luminous color of the light-emitting module **15**, or controlling a working mode of the light-emitting module **15**. The working mode is not limited to an automatic change in brightness of the light, or an automatic change in a color of the light. For example, an ambient light sensor and a timer may be integrated with the circuit board. Therefore, automatic turning on or off in a preset period of time may be implemented by detecting intensity of ambient light in the evening in combination with the timer. In some implementations, the light-emitting module **15** may be integrated with the circuit board.

Referring to FIG. **4**, a light pole assembly **13** includes a mounting assembly **131** connected to the ground plug pin **11** and a protective cover **133** connected to the mounting assembly **131**. The light-emitting module **15** and the functional member **17** are disposed in the mounting assembly **131**. The protective cover **133** has an open state and a closed state relative to the mounting assembly **131**. The functional member **17** is in a telecommunication connection or a communication connection to the control circuit. The functional member **17** is not limited to a mechanical button or a touch button, and may further be a trigger structure or an adjustment structure for providing another function. The present disclosure uses a button as an example for description. When the protective cover **133** is in the open state, as shown in FIG. **2**, the functional member **17** is exposed, and a user may press the button to control the operating state, the luminous intensity, the luminous color, the working mode of the light-emitting module **15**. When the protective cover **133** is in the closed state, as shown in FIG. **1**, the protective cover **133** obscures the functional member **17**, to avoid an accidental contact generated because the functional member **17** is exposed. Certainly, the protective cover **133** may further enable a waterproof function for the functional member **17**, to ensure working reliability and service life of the functional member **17**. Certainly, in some implementations, the protective cover **133** may further prevent the functional member **17** from being affected by dust, that is, dustproof effect.

In other implementations, the functional member **17** may be integrated with a wireless signal receiving module, for example, an infrared signal receiving module, and integrated with a short-range wireless signal receiving module, for example, a 433 MHz signal receiving module or a 2.4G signal receiving module. When the protective cover **133** is in the closed state, the protective cover **133** and the mounting assembly **131** enclose the functional member **17** in isolated space. Therefore, the protective cover **133** may block a close-range wireless communication signal or enable a close-range wireless communication signal to be absorbed or reflected, and the close-range wireless communication signal is attenuated until the functional member **133** is not triggered, thereby providing anti-trigger effect in the closed state and implementing normal triggering in the open state.

In some implementations, the protective cover **133** is detachably connected to the light pole assembly **13**. Specifically, in the embodiment 1 of the present disclosure, the mounting assembly **131** includes a mounting base **1311** and a middle pole **1313**. The light-emitting module **15** and the functional member **17** are disposed in the mounting base **1311**, the middle pole **1313** is in a long tube shape, and has one end connected to the ground plug pin **11** and the other end connected to the mounting base **1311**, and the protective cover **133** is connected to the mounting base **1311**. In some implementations, the middle pole **1313** is in a hollow tube shape, the ground plug pin **11** is detachably connected to the middle pole **1313**, a wire hole **11*a*** is provided at a junction, and the wire hole **11*a*** is configured to allow a cable to pass through. The cable can pass through the middle pole **1313** and extend to the light-emitting module **15** of the mounting base **1311**, to be in a telecommunication connection to the light-emitting module **15**. The cable may be used for an external power supply, to supply power to the light-emitting module **15**. In other implementations, the middle pole **1313** may be in other shapes such as a square tube shape.

Still referring to FIG. **3** and FIG. **4**, the mounting base **1311** includes a base plate **13111**, a carrier member **13113**, and a light guide member **13115** connected between the base plate **13111** and the carrier member **13113**. The middle pole **1313** is connected to the base plate **13111**, the light-emitting module **15** and the functional member **17** are disposed in the carrier member **13113**, and the protective cover **133** is connected to the carrier member **13113**. For example, in the implementation in FIG. **2**, the protective cover **133** is connected to the carrier member **13113** in a threaded manner. For example, the protective cover **133** has an internal thread, and the carrier member **13113** has an external thread that is fitted with the internal thread of the protective cover **133**, to implement the screwing between the internal thread and the external thread. The protective cover **133** may be easily unscrewed from the carrier member **13113** without the use of other tools, thereby exposing the functional member **17** and facilitating an operation by the user. After the operation is completed, the user can conveniently mount the protective cover **133** in the carrier member **13113**, thereby preventing an accidental contact and providing protection for the functional member **17**. In other implementations, a side of the protective cover **133** may be rotatably connected to the carrier member **13113**. Therefore, the protective cover **133** may be in the open state or the closed state relative to the carrier member **13113**. In other implementations, the protective cover **133** may be clamped in the carrier member **13113**, and may also be conveniently opened and closed. Therefore, the protective cover **133** may be in the open state or the closed state. This is convenient for the operation of the user.

The carrier member **13113** may be used to provide mounting space for the circuit board and light-emitting module **15** of the plug-in ground light **10**, limit and protect the circuit board and light-emitting module **15** of the plug-in ground light **10**. The carrier member **13113** may be a metal member, or a non-metallic member with relatively low light transmittance, for example, a plastic member or a ceramic member. A material of the base plate **13111** may be the same as or not the same as that of the carrier member **13113**. In the implementations in FIG. **3** and FIG. **4**, the light guide member **13115** is in a cylindrical shape and partially extends in the carrier member **13113**. The carrier member **13113** encloses at least a part of the light guide member **13115**, to reliably limit the light guide member **13115** into the carrier member **13113**. Due to shading effect of the carrier member **13113**, leakage of light from a connection position of the light guide member **13115** and the carrier member **13113** is reduced. A material of the light guide member **13115** may be plastic or glass with suitable light transmittance. Light emitted from the light-emitting module **15** may be emitted to the light guide member **13115**, and a propagation direction of the light may be changed via the light guide member **13115**, for example, by dispersing the light, thereby expanding an irradiation range of the light-emitting module **15**.

Further, referring to FIG. **4**, a through hole **13115***a* is axially provided on the light guide member **13115**, the through hole **13115***a* extends from one end of the light guide member **13115** to the other end of the light guide member **13115** along a center shaft of the light guide member **13115**, and the through hole **13115***a* communicates with an interior of the middle pole **1313** and configured to allow the cable to pass through. This structure prevents the cable from being exposed outside, to protect the cable.

Further, one end, away from the carrier member **13113**, of the light guide member **13115** is recessed towards a side on which the carrier member **13113** is located, to form a groove **13115***b*, and the groove **13115***b* communicates with the through hole **13115***a*. The groove **13115***b* may be used to avoid a connection part of the base plate **13111** and the middle pole **1313**, to accommodate the connection part of the base plate **13111** and the middle pole **1313** into the groove **13115***b*, thereby improving structural compactness, and enabling the plug-in ground light **10** to have a simple appearance.

Further, the mounting base **1311** may include a sealing member **13117**. The sealing member **13117** is in an annulus shape and circumferentially provided with an accommodation groove **13117***a*. A material of the sealing member **13117** may be silica gel or rubber. A convex ring **13115***c* matching the accommodation groove **13117***a* is disposed at each of two opposite ends of the light guide member **13115**. There may be two sealing members **13117**. One sealing member **13117** is configured to seal a gap between the light guide member **13115** and the base plate **13111**, and the other sealing member **13117** is configured to seal a gap between the carrier member **13113** and the light guide member **13115**, to enable the mounting base **1311** of the plug-in ground light **10** to have good waterproof and dustproof performance. Further, in some implementations, a side, facing the base plate **13111**, of the carrier member **13113** and a side, facing the carrier member **13113**, of the base plate **13111** may be recessed, to form a recessed groove **1311***a*, to accommodate the sealing member **13117** in the recessed groove **1311***a*, thereby improving reliability of limiting the sealing member **13117** in the carrier member **13113** and the base plate **13111**, and improving structural compactness of the plug-in ground light **10**.

Further, referring to FIG. **3** and FIG. **4**, the mounting base **1311** may include a reflective sheet **13119**, and the reflective sheet **13119** is disposed on a side, facing the light-emitting module **15**, of the base plate **13111**. The reflective sheet **13119** has a surface with relatively high reflectivity, and the surface with relatively high reflectivity faces the light-emitting module **15**, to reflect a part of the light emitting to the base plate **13111** by the light guide member **13115** back to the light guide member **13115**, thereby improving a utilization rate of light energy. The reflective sheet **13119** may be an aluminum foil, or a non-metallic sheet coated with a thin reflective film.

Further, in some implementations, two opposite ends of the middle pole **1313** may be connected to the base plate **13111** and the ground plug pin **11** respectively in a threaded manner. In this implementation, convenience for assembling the middle pole **1313**, the ground plug pin **11**, and the base plate **13111** is improved, and convenience for transporting, disassembling, and maintaining the plug-in ground light **10** is improved, to enable the user to conveniently assemble the plug-in ground light **10**, and conveniently disassemble and replace the plug-in ground light **10** in case of a problem. Similarly, the base plate **13111** and the carrier member **13113** may be detachably connected to the light guide member **13115** respectively, to improve convenience for assembling the base plate **13111**, the carrier member **13113**, and the light guide member **13115**, and improve convenience for transporting, disassembling, and maintaining the plug-in ground light **10**, thereby enabling the user to conveniently assemble the plug-in ground light **10**, and conveniently disassemble and replace components of the plug-in ground light **10** in case of a problem. In particular, due to this structure, the user may conveniently replace the base plate **13111**, the light guide member **13115**, and the carrier member **13113** in different shapes, thereby expanding usage scenes.

The plug-in ground light **10** includes the ground plug pin **11**, the light pole assembly **13**, the light-emitting module **15**, and the functional member **17**. The light pole assembly **13** includes the mounting assembly **131** connected to the ground plug pin **11** and the protective cover **133** connected to the mounting assembly **131**. The protective cover **133** has the open state and the closed state relative to the mounting assembly **131**. The light-emitting module **15** is disposed in the mounting assembly **131**, and the functional member **17** is disposed in the mounting assembly **131**. In addition, when the protective cover **133** is in the open state, the functional member **17** is exposed. When the protective cover **133** is in the closed state, the protective cover **133** obscures the functional member **17**. The functional member **17** is configured to control the at least one of the operating state, the luminous intensity, the luminous color, and the operating mode of the light-emitting module **15**. The protective cover **133** may avoid an accidental contact generated because the functional member **17** is exposed, and may enable a waterproof function for the functional member **17**, to ensure working reliability and service life of the functional member **17**. Certainly, in some implementations, the protective cover **133** may further prevent the functional member **17** from being affected by dust, that is, dustproof effect. When the protective cover **133** is in the open state, the user can conveniently adjust the operating state, the luminous intensity, the luminous color, the working mode of the light-emitting module **15**, thereby improving flexibility of controlling the plug-in ground light, and expanding the use scenes of the plug-in ground light **10**. For example, the plug-in ground light **10** is used to create dynamic lighting effect during a courtyard party.

Embodiment 2

Referring to FIG. **5**, FIG. **6**, and FIG. **7**, Embodiment 2 of the present disclose provides a plug-in ground light **20**. The plug-in ground light **20** includes a ground plug pin **21**, a light pole assembly **23**, a light-emitting module **25**, and a functional member **27**. The light pole assembly **23** is connected to the ground plug pin **21**, and the light-emitting module **25** and the functional member **27** are disposed in the light pole assembly **23**. One end of the ground plug pin **21** is relatively sharp and may be inserted into the ground, thereby reliably fixing the plug-in ground light **20** on the ground for providing lighting.

In addition, referring to FIG. **8**, similar to the embodiment 1, a light pole assembly **23** also includes a mounting assembly **231** connected to the ground plug pin **21** and a protective cover **233** connected to the mounting assembly **231**. The light-emitting module **25** and the functional member **27** are disposed in the mounting assembly **231**. The protective cover **233** has an open state and a closed state relative to the mounting assembly **231**. The functional member **27** is in a telecommunication connection or a communication connection to the control circuit. The functional member **27** is not limited to a mechanical button or a touch button and may further be a trigger structure or an adjustment structure for providing another function. The embodiment 2 of the present disclosure takes a button as an example for description. When the protective cover **233** is in the open state, as shown in FIG. **6**, the functional member **27** is exposed, and a user may press the button to control the operating state, the luminous intensity, the luminous color, the working mode of the light-emitting module **25**. When the protective cover **233** is in the closed state, as shown in FIG. **5**, the protective cover **233** obscures the functional member **27**, to avoid an accidental contact generated because the functional member **27** is exposed, and may enable a waterproof function for the functional member **27**, to ensure working reliability and service life of the functional member **27**. Certainly, in some implementations, the protective cover **233** may further prevent the functional member **27** from being affected by dust, that is, dustproof effect.

In some implementations, the protective cover **233** is detachably connected to the light pole assembly **23**. Specifically, in the embodiment 2 of the present disclosure, the mounting assembly **231** may also include a mounting base **2311** and a middle pole **2313**. The light-emitting module **25** and the functional member **27** are disposed in the mounting base **2311**. The middle pole **2313** is in a hollow tube shape, the ground plug pin **21** is detachably connected to the middle pole **2313**, a wire hole **21***a* is provided at a junction, and the wire hole **21***a* is configured to allow a cable to pass through. The cable can pass through the middle pole **2313** and extend to the light-emitting module **25** of the mounting base **2311**, to be in a telecommunication connection to the light-emitting module **25**. The cable may be used for an external power supply, to supply power to the light-emitting module **25**. In other implementations, the middle pole **2313** may be in other shapes such as a square tube shape, which is not described herein again.

Referring to FIG. **7** and FIG. **8**, the mounting base **2311** includes a base plate **23111**, a carrier member **23113**, and a light guide member **23115** connected between the base plate **23111** and the carrier member **23113**. The middle pole **2313** is connected to the base plate **23111**, the light-emitting module **25** and the functional member **27** are disposed in the carrier member **23113**, and the protective cover **233** is connected to the carrier member **23113**. For example, in implementations in FIG. **7** and FIG. **8**, the protective cover **233** is substantially tapered, has a relatively large holding area, and is connected to the carrier member **23113** in a threaded manner. The protective cover **233** may be easily unscrewed from the carrier member **23113** without the use of other tools, thereby exposing the functional member **27** and facilitating an operation by the user. After the operation is completed, the user can conveniently mount the protective cover **233** in the carrier member **23113**, thereby providing a waterproof function and protection for the functional member **27**. In other implementations, the protective cover **233** may be replaced with a structure in which a side is rotatably connected to the carrier member **23113**. Therefore, the protective cover **233** may be in the open state or the closed state relative to the carrier member **23113**. This is not described herein again.

The carrier member **23113** may be used to provide mounting space for the circuit board and light-emitting module **25** of the plug-in ground light **20**, limit and protect the circuit board and light-emitting module **25** of the plug-in ground light **20**. The carrier member **23113** may be a metal member, or a non-metallic member with relatively low light transmittance, for example, a plastic member or a ceramic member. In the implementations FIG. **7** and FIG. **8**, a cross-section of the light guide member **23115** is substantially in a shape of a trapezoid with a curved side, and the protective cover **233** is covered on the light guide member **23115**. Specifically, the light guide member **23115** includes a first end **23115***a* and a second end **23115***b* that are opposite to each other, outer contours of the first end **23115***a* and the second end **23115***b* are circular, and an outer diameter of the first end **23115***a* is larger than an outer diameter of the second end **23115***b*. The protective cover **233** is disposed on the carrier member **23113**, and an outer diameter of the protective cover **233** is larger than the outer diameter of the first end **23115***a*. The base plate **23111** is connected to the second end **23115***b*, the carrier member **23113** is connected to the first end **23115***a*, and in an axial direction of the middle pole **2313**, an outer diameter of the light guide member **23115** gradually decreases from the first end **23115***a* to the second end **23115***b*.

A material of the light guide member **23115** may be plastic or glass with suitable light transmittance. Light emitted from the light-emitting module **25** may be emitted to the light guide member **23115**, and a propagation direction of the light may be changed via the light guide member **23115**, for example, by dispersing the light, thereby expanding an irradiation range of the light-emitting module **25**. A gain pattern may further be disposed on a circumferential surface of the light guide member **23115**, thereby changing light-emitting effect and expanding the use scenes of the plug-in ground light **20**.

Differences with the embodiment 1 lie in that shapes of the base plate **23111**, the light guide member **23115**, the carrier member **23113**, and the protective cover **233** are changed significantly, the base plate **23111**, the carrier member **23113**, and the light guide member **23115** form the mounting cavity **2311***a*, and the light-emitting module **25** is disposed in the mounting cavity **2311***a*. An inner surface of the light guide member **23115** may be a conical surface or a parabolic surface. Therefore, light in the mounting cavity **2311***a* is reflected via the inner surface of the light guide member **23115**, to implement diffusing effect and improve a utilization rate of light energy. For the function and working principle of the above components, refer to the embodiment 1. This is not described in the present disclosure.

In the embodiment 2 of the present disclosure, the two opposite ends of the middle pole **2313** may further be detachably connected to the base plate **23111** in a one-to-one correspondence manner, to improve convenience for assembling the middle pole **2313**, the ground plug pin **21**, and the base plate **23111**, and improve convenience for transporting, disassembling, and maintaining the plug-in ground light **20**, thereby enabling the user to conveniently assemble the plug-in ground light **20**, and conveniently disassemble and replace the plug-in ground light **20** in case of a problem. Similarly, the base plate **23111** may also be detachably connected to the light guide member **23115**, the light guide member **23115** may also be detachably connected to the carrier member **23113**, the carrier member **23113** may also be detachably connected to the protective cover **233**, to improve convenience for assembling the base plate **23111**, the carrier member **23113**, and the light guide member

23115, and improve convenience for transporting, disassembling, and maintaining the plug-in ground light 20, thereby enabling the user to conveniently assemble the plug-in ground light 20, and conveniently disassemble and replace the plug-in ground light 20 in case of a problem. In particular, due to this structure, the user may conveniently replace the base plate 23111, the light guide member 23115, and the carrier member 23113 in different shapes, thereby expanding usage scenes. This is not described herein again.

The technical features of the foregoing embodiments can be employed in arbitrary combinations. For brevity of description, not all possible combinations of the technical features of the foregoing embodiments are described. However, the combinations of the technical features should be construed as falling within the scope described in this specification as long as there is no contradiction in the combinations.

The above described are merely several implementations of the present disclosure. Although the embodiments are described specifically and in detail, they should not be construed as a limitation to the patent scope of the present disclosure. It should be noted that those of ordinary skill in the art can further make several variations and improvements without departing from the concept of the present disclosure, and all of these fall within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be subject to the protection scope defined by the claims.

What is claimed is:

1. A plug-in ground light, comprising:

a ground plug pin;

a light pole assembly, comprising a mounting assembly connected to the ground plug pin and a protective cover connected to the mounting assembly, wherein the protective cover has an open state and a closed state relative to the mounting assembly;

a light-emitting module, disposed in the mounting assembly; and

a functional member, disposed in the mounting assembly, wherein when the protective cover is in the open state, the functional member is exposed; when the protective cover is in the closed state, the protective cover obscures the functional member; and the functional member is configured to control at least one of an operating state, luminous intensity, a luminous color, and an operating mode of the light-emitting module.

2. The plug-in ground light according to claim **1**, wherein the protective cover is detachably connected to the light pole assembly.

3. The plug-in ground light according to claim **1**, wherein the protective cover is rotatably connected to the light pole assembly.

4. The plug-in ground light according to claim **1**, wherein the light pole assembly comprises a mounting base and a middle pole, the light-emitting module and the functional member are disposed in the mounting base, the middle pole has one end connected to the ground plug pin and the other end connected to the mounting base, and the protective cover is connected to the mounting base.

5. The plug-in ground light according to claim **4**, wherein the middle pole is in a hollow tube shape, the ground plug pin is detachably connected to the middle pole, a wire hole is provided at a junction of the ground plug and the middle pole, the wire hole is configured to allow a cable to pass through, and the cable passes through the middle pole and extends to the light-emitting module of the mounting base.

6. The plug-in ground light according to claim **5**, wherein the mounting base comprises a base plate, a carrier member, and a light guide member connected between the base plate and the carrier member, the middle pole is connected to the base plate, the light-emitting module and the functional member are disposed in the carrier member, and the protective cover is connected to the carrier member.

7. The plug-in ground light according to claim **6**, wherein two opposite ends of the middle pole are connected to the mounting base and the ground plug pin respectively in a threaded manner.

8. The plug-in ground light according to claim **6**, wherein the base plate and the carrier member are detachably connected to the light guide member.

9. The plug-in ground light according to claim **6**, wherein the base plate, the carrier member, and the light guide member form a mounting cavity, and the light-emitting module is disposed in the mounting cavity.

10. The plug-in ground light according to claim **9**, wherein the light guide member comprises a first end and a second end that are opposite to each other, outer contours of the first end and the second end are circular, an outer diameter of the first end is larger than an outer diameter of the second end, the base plate is connected to the second end, the carrier member is connected to the first end, and in an axial direction of the middle pole, an outer diameter of the light guide member gradually decreases from the first end to the second end.

11. The plug-in ground light according to claim **10**, wherein the protective cover is disposed on the carrier member, and an outer diameter of the protective cover is larger than the outer diameter of the first end.

12. The plug-in ground light according to claim **6**, wherein a through hole is axially provided on the light guide member, the through hole extends from one end of the light guide member to the other end of the light guide member, the through hole communicates with an interior of the middle pole and configured to allow the cable to pass through.

13. The plug-in ground light according to claim **12**, wherein one end, away from the carrier member, of the light guide member is recessed towards a side on which the carrier member is located, to form a groove, and the groove communicates with the through hole.

14. The plug-in ground light according to claim **12**, wherein the mounting base comprises two sealing members, the sealing member is in an annulus shape and circumferentially provided with an accommodation groove, a convex ring matching the accommodation groove is disposed at each of two opposite ends of the light guide member, one sealing member is configured to seal a gap between the light guide member and the base plate, and the other sealing member is configured to seal a gap between the carrier member and the light guide member.

15. The plug-in ground light according to claim **12**, wherein the mounting base comprises a reflective sheet, and the reflective sheet is disposed on a side, facing the light-emitting module, of the base plate.

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